Michael Wieck-Sosa

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EDUCATION

Carnegie Mellon University | PhD in Statistics | Advisors: Aaditya Ramdas and Cosma Shalizi *Sept. 2022–Present*

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by department faculty to one PhD student per year
- Thesis topics: generative models and time series | Committee: Aaditya Ramdas, Cosma Shalizi, Yuting Wei, Wei Biao Wu

University of Illinois at Urbana-Champaign | MS in Statistics *May 2022*

- GPA: 3.95/4.00 | Awards: two years of assistantships (tuition waiver and stipend) | GRE: 170/170 Q, 163/170 V, 4.5/6.0 W

Fordham University | BS in Mathematics with minors in Computer Science and Economics *May 2020*

- GPA: 3.77/4.00 | Awards: *magna cum laude* | Research: deep learning-based tracking system for an endangered toad species

PROGRAMMING LANGUAGES AND TOOLS

- Python expert: extensive experience with NumPy, SciPy, pandas, scikit-learn, statsmodels, and PyTorch
- R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
- C++ proficient: used in Data Structures and Algorithms courses
- SQL, Git, Bash expert: extensive experience with data management, version control, and Linux

COURSEWORK

- **Statistics**: Advanced Statistical Theory, Mathematical Statistics, Time Series Analysis, Regression, Computational Statistics
- **Computer Science**: Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
- **Mathematics**: Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids and Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

SELECTED RESEARCH ASSISTANTSHIPS AND INTERNSHIPS

Carnegie Mellon University | NSF Grant (PI: Cosma Shalizi) | Research Assistant *May 2024–Present*

- Developed theory, methods, and software for fitting complex scientific models using nonconvex optimization in Python
- Trained generative sequence models with transformers and other deep neural network architectures on GPUs using PyTorch

National Center for Supercomputing Applications | Great Lakes to Gulf Project | Research Assistant *Sept. 2020–May 2022*

- Built confidence bands for chemical concentrations and fluxes in water over time at 1,000+ sites via parallel computing in R

MIT Lincoln Laboratory | Group 38 (Sensor Technology) | Research Intern *May 2021–July 2021*

- Tracked objects in outer space and conducted simulation studies to compare different tracking methods using MATLAB

University of Illinois at Urbana-Champaign | Computer Science Department | Research Assistant *Jan. 2021–May 2021*

- Modeled dynamics of misinformation spreading across multiple social media platforms using Hawkes processes with Python

TEACHING ASSISTANTSHIPS

- MS in Computational Finance program at CMU: Deep Learning, Natural Language Processing, Simulation Methods for Option Pricing, Financial Time Series Analysis, Financial Data Science I and II, Machine Learning Capstone Project (x2)
- BS in Statistics/StatML and MS in Data Science programs at CMU: Time Series, Advanced Data Analysis

SELECTED PROFESSIONAL SERVICE

- Paper reviewer | Journal of the Royal Statistical Society, Series B (Statistical Methodology) *2026*
- Committee chair | CMU Statistics PhD student committee for organizing career events *2024–2025*

SELECTED TALKS AND POSTERS

- Invited talk on deep Granger causality with transformers | CFE-CMStatistics 2026 | HTW Berlin *2026*
- Invited talk on conditional independence testing | IWSM 2026 | American University *2026*
- Contributed poster on conditional independence testing | NBER-NSF Time Series Conference | UPenn *2024*